Statistical Assurance Regions for Predictive Models Under Joint Distribution Shift and Missingness

Faustus Domebale Maale FDMAALE@AGGIES.NCAT.EDU
Applied Science and Technology Ph.D. Program
College of Science and Technology
North Carolina A&T State University, Greensboro, NC, USA

Hui Yi HYI@NCAT.EDU
Department of Mathematics & Statistics
North Carolina A&T State University, Greensboro, NC, USA

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Abstract

Deployment changes in populations and measurement availability can undermine predictive performance, making test-set averages insufficient for identifying conditions that meet a specified risk tolerance. We define statistical assurance regions (SARs) over prespecified environments jointly modeling missingness and distribution shift, with risk defined as expected bounded loss. We estimate conservative inner SARs contained in the true region with probability at least $1 - \alpha$: in finite samples using exact-binomial and bounded-loss procedures, and asymptotically using an optional Gaussian multiplier band. Simulations show that the multiplier increased assurance-region recall by 0.016–0.092 over exact Bonferroni but undercovered for binary loss, so exact procedures are the default. Power analysis yields exact sample-size planning. In Wisconsin Diagnostic Breast Cancer (WDBC) stress tests, exact Bonferroni certified 15 of 36 logistic-regression conditions at a misclassification-risk tolerance of 0.08. Mortality prediction used 35,114 admissions from the Medical Information Mart for Intensive Care (MIMIC-IV) database. A loose misclassification-risk tolerance certified all 60 eligible conditions, including an always-survive classifier; requiring sensitivity of at least 0.80 certified only three under exact Bonferroni. Approximate temporal domains preclude claims of measured temporal shift. SARs map deployment conditions supported by simultaneous risk guarantees, conditional on the loss, tolerance, environment family, and sampling design.

Keywords: deployment assurance, distribution shift, missing data, simultaneous inference, risk control

1 Introduction

An acceptable test-set average does not specify the deployment conditions under which a trained machine learning model can meet a performance requirement stated as an upper limit on its risk, such as a maximum misclassification or false-negative rate (Quiñonero Candela et al., 2009; Recht et al., 2019; Koh et al., 2021; Zhou et al., 2023). The need to examine distribution changes extends across several application areas. In healthcare, patient populations, clinical practices, and measurement availability can change between model development and deployment (Finlayson et al., 2021; Groenwold, 2020). Autonomous-driving

perception systems face changing weather and illumination, including fog, rain, snow, and nighttime conditions (Sakaridis et al., 2021). Researchers have also studied shifts across camera locations in wildlife monitoring, acquisition locations and sensors in agricultural monitoring, geographical regions and time periods in satellite-image analysis, and countries in poverty estimation, as well as molecular-property prediction across chemical scaffolds, cell-image classification across experimental batches, sentiment analysis across reviewers, toxicity classification across demographic-identity mentions, and code completion across software repositories (Koh et al., 2021). A useful validation report would identify which explicitly modeled deployment conditions have a simultaneous upper confidence bound on the model's misclassification probability below a specified tolerance, rather than merely a low observed misclassification rate. For the class-sensitive analysis, the corresponding target is the false-negative probability among positive cases.

We study this reporting problem for a predictive pipeline that is fixed before the assurance data are examined. Given environments $e_1, \ldots, e_K$, predictive risks $R_1, \ldots, R_K$, and a tolerance τ, the target is $\mathcal{A}_\tau = \{e_k : R_k \leq \tau\}$. A simultaneous upper band $(U_1, \ldots, U_K)$ produces the inner estimate $\widehat{\mathcal{A}}_\tau = \{e_k : U_k \leq \tau\}$, and we say that the environments in $\widehat{\mathcal{A}}_\tau$ are certified. The word is used only in this statistical sense; it does not denote regulatory approval or clinical safety. For example, upper bounds $(.07, .09, .13)$ at tolerance .10 certify the first two environments. The third remains uncertified even if its point estimate is below .10; it has not thereby been shown to be unacceptable. We call $\mathcal{A}_\tau$ the statistical assurance region (SAR) and $\widehat{\mathcal{A}}_\tau$ its inner estimate. Figure 1 summarizes the framework.

The inferential devices underlying this construction are established. Exact binomial limits, bounded-sum inequalities, multiple testing, and Gaussian multiplier approximation already control joint error (Clopper and Pearson, 1934; Hoeffding, 1963; Holm, 1979; Chernozhukov et al., 2013), and Learn then Test (LTT) can index its hypotheses by deployment environments (Angelopoulos et al., 2025). Naming the resulting set does not create a new inferential theorem. We therefore treat SARs as an applied methodology and implement them in an open-source procedure, which we call MiSCert (missingness-and-shift certification), that links an explicit environment design, its sampling assumptions, the accounting of uncertainty, and an interpretable set-valued report.

The paper makes five contributions. First, it separates finite-sample risk control from asymptotic dependence-aware inference and states when each is appropriate. Second, it gives the row-level construction that justifies reusing observations under several transformations, including a missingness generator whose missing-at-random (MAR) mechanism depends only on always-observed covariates. Third, it evaluates the procedures on nested fitted-model experiments in which the reference risks are themselves uncertain. Fourth, it derives a closed-form certification probability for the exact procedure that supports sample-size planning, and explains why Gaussian bands undercover in large families even at large sample sizes. Fifth, it applies the procedure to credentialed MIMIC-IV v3.1 data using admission-aligned approximate temporal domains, one admission per patient, native and added missingness, two learned models, and an always-survive classifier, under a misclassification target and a class-sensitive false-negative target. All tables and figures are generated by one shared implementation and executable notebook.

The results do not support universal use of the multiplier band. It can enlarge the certified region when losses across environments are strongly dependent, but its simultaneous

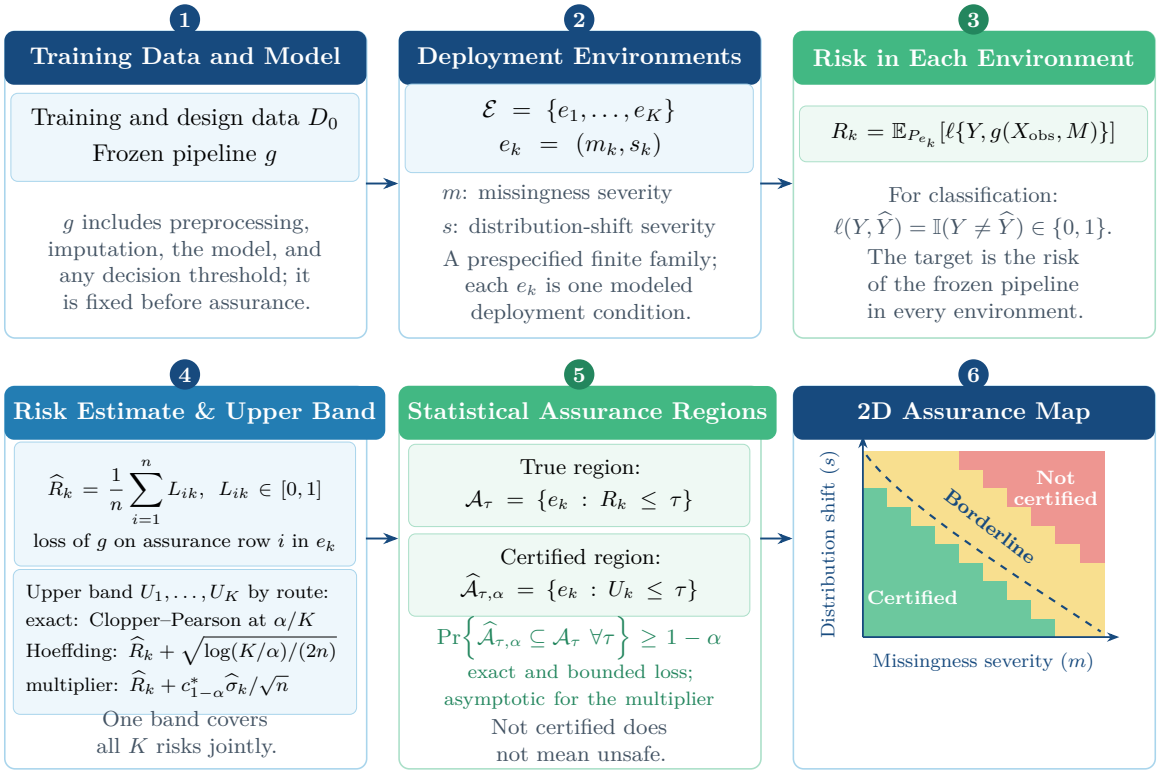


Figure 1: Overview of the statistical assurance framework. A frozen pipeline g is evaluated over a prespecified family of K deployment environments indexed by missingness severity m and distribution-shift severity s. One simultaneous upper band $(U_1, \ldots, U_K)$, from an exact binomial, bounded-loss, or multiplier route, defines the certified inner region $\widehat{\mathcal{A}}_{\tau,\alpha}$. The exact and bounded-loss routes give containment for all τ with probability at least $1 - \alpha$ in finite samples; the multiplier route gives it asymptotically. In the stylized map, green cells are certified, yellow cells have estimated risk at most τ but are not certified, and red cells are not certified; neither uncertified category is a proof that the environment is unsafe.

coverage was 0.928 in the primary calibration (95% Monte Carlo interval $[0.902, 0.948]$) and fell to 0.780 in a rare-event design, both below the nominal 0.95. We therefore use exact Clopper–Pearson Bonferroni or exact Holm as the default procedures for binary losses and report the multiplier as a secondary analysis whose approximation must be validated rather than assumed. Among valid procedures, the margin between each risk and the tolerance and the size of the assurance sample, rather than the choice of procedure, determine how much is certified. In the MIMIC-IV analysis the binding constraint for the false-negative target was the number of deaths in each domain.

Section 1.1 reviews related work. Section 2 defines the target and the sampling designs, and Section 3 presents the inferential procedures, the extension to continuous environment families, and sample-size planning. Section 4 reports simulation studies of validity, power,

fitted pipelines, and missingness mechanisms. Section 5 analyzes the WDBC and MIMIC-IV data, and Section 6 discusses practical use and limitations. Additional results, computational details, and the provenance of each reported result are given in the appendices.

1.1 Related Work

Risk-controlling prediction sets and LTT provide model-agnostic calibration and testing frameworks (Bates et al., 2021; Angelopoulos et al., 2025). In our finite-family setting, the hypotheses $H_{0k} : R_k \geq \tau$ are an immediate application of LTT, so the environment-indexed assurance report is an operational interpretation of that framework rather than a new class of guarantees. Simultaneous confidence bands also have a long history outside high-dimensional Gaussian maxima; Degras (2011), for example, studies bands for functional regression. Here the coordinates are prespecified environment-specific predictive risks, and the focus is on their construction and sampling design rather than on a new band theorem.

A second line of work predicts or adapts performance under shift. Average thresholded confidence (ATC) predicts target accuracy from source-calibrated confidence scores and unlabeled target observations (Garg et al., 2022). Because its inputs and objective differ from labeled certification, we evaluate it by its risk-prediction error rather than by comparing it with a familywise procedure. Weighted conformal prediction addresses covariate shift through likelihood-ratio information under its stated conditions (Tibshirani et al., 2019); it does not by itself give a simultaneous region over an arbitrary family of missingness interventions. Adaptation to missingness shift changes the predictor for target conditions (Zhou et al., 2023), and an adapted pipeline could then be fixed and assessed with the procedure studied here.

Generalization bounds and robust learning address related but distinct questions. PAC-Bayes analysis controls the generalization of distributions over predictors through complexity terms (Alquier, 2024), and distributionally robust optimization changes the training objective to improve performance over an uncertainty set (Duchi and Namkoong, 2021). Our procedure instead conditions on a fitted pipeline and uses independent labeled assurance observations. It does not claim to dominate these approaches, and it does not transport labels to unobserved deployment populations without additional assumptions.

Finally, region-based assurance has precedents in engineering. Dixit et al. (2025) couple conformal calibration of perception to a safe planner and evaluate navigation safety; their guarantee concerns the resulting control system, whereas a risk threshold alone says nothing about downstream control or clinical harm. Torens et al. (2025) monitor an operational design domain for a machine learning component, and Hammar et al. (2026) learn a network's safe operating region online through causal observation and intervention, with a guarantee that the learned region is safe with a specified probability. Our analysis is offline and does not expand the region actively; in our setting, uncertainty in the average predictive loss determines an inner set within an explicitly supplied family.

2 Problem Formulation and Sampling Design

Let D_0 denote all training and design information, including preprocessing estimates, learner randomization, a pilot sample if one is used, and the selected finite environment family $\mathcal{E}(D_0) = \{e_1, \ldots, e_K\}$. We condition on D_0 throughout unless stated otherwise. The fitted

pipeline g includes imputation, transformations, prediction, and any fixed decision threshold. For a loss ℓ with values in $[0,1]$, define

$$R_g(e) = \mathbb{E}_{P_e}[\ell(Y, g(X_{\text{obs}}, M))], \qquad \mathcal{A}_\tau(g) = \{e \in \mathcal{E}(D_0) : R_g(e) \le \tau\}, \tag{1}$$

where X_{obs} denotes the observed covariates and M the missingness pattern. The environment law P_e must be specified. Synthetic perturbation experiments establish results only for their artificial environment laws, not automatically for a real hospital or calendar period. The family $\mathcal{E}(D_0)$ is a scientific input: it should start from a specified loss and decision threshold, use interpretable axes with externally motivated ranges, and distinguish changes in the population from changes in measurement. Axes that are omitted remain outside the scope of the report.

2.1 Shared Transformations of One Sample

Lemma 1 (Shared transformations) *Conditional on D_0, let (Z_i, V_i) be i.i.d. across i and independent of D_0, and let $h_k(\cdot; D_0)$ be fixed measurable maps into $[0,1]$. Then $\mathbf{L}_i = (h_1(Z_i, V_i; D_0), \ldots, h_K(Z_i, V_i; D_0))$ are i.i.d. bounded random vectors. No independence across coordinates is required.*

Proof For any measurable sets $B_1, \ldots, B_n$, the events $\{\mathbf{L}_i \in B_i\}$ are measurable with respect to the respective independent rows (Z_i, V_i), conditional on D_0, so their joint conditional probability factors. The same row law and fixed maps give the same vector distribution for each i. ■

Shared masks, noise, or outcome uniforms within a row are therefore compatible with simultaneous inference, and perfectly correlated coordinates are permitted. In contrast, normalizing missingness probabilities by the mean of the assurance batch makes the transformation depend on other rows and is not covered by Lemma 1; our generators therefore use fixed population-calibrated intercepts. Repeated patients, temporal dependence, or cluster sampling require an analysis beyond this i.i.d. argument, and conditioning on training data does not remove dependence among assurance rows.

2.2 Natural Groups and Independent Samples

For a master sample of independent subjects, let G_{ig} indicate that subject i belongs to a prespecified domain or subgroup g, and let j index the stress or model conditions. With $n_g = \sum_i G_{ig}$ and $\widehat{p}_g = n_g/n$, define

$$\widehat{R}_{g,j} = \frac{\sum_i G_{ig} L_i^{(j)}}{n_g}, \qquad \widehat{\psi}_{i,g,j} = \frac{G_{ig}}{\widehat{p}_g}(L_i^{(j)} - \widehat{R}_{g,j}). \tag{2}$$

Common row multipliers applied to these influence estimates preserve both overlapping-group and within-subject dependence. Fixed-dimensional asymptotic ratio inference additionally requires positive group probabilities and nondegenerate influence variances. Small or empty groups remain in the report and cannot be certified under a prespecified minimum-size rule. For exact binomial inference, we condition on the group membership indicators: under i.i.d.

master rows and a fixed group definition, the binary losses selected by a group have the group-conditional Bernoulli law, and a union bound covers all groups and stress conditions despite overlap. When environments instead have genuinely independent samples of unequal size, the procedure accepts separate loss vectors; it neither truncates all samples to the shortest length nor pairs unrelated subjects by row position. Validity then concerns the represented environment populations, not unobserved sites.

2.3 Class-Conditional Risks

Some targets, such as the false-negative risk $\Pr\{\widehat{Y} = 0 \mid Y = 1, e\}$, condition on the label. Conditional on all pairs (Y_i, G_{ig}) and on D_0, the rows are independent and every row in the cell $\{Y = 1, G = g\}$ has the same law. The indicator $\mathbb{I}\{\widehat{Y}_i = 0\}$ for those rows is therefore Bernoulli with the class-conditional risk of the cell, and the exact bound (3) below applies with n_k equal to the number of positive cases in the cell. An eligibility rule that depends only on the counts of (Y_i, G_{ig}), such as a minimum number of positive cases, is measurable with respect to the conditioning variables and does not invalidate the bound. The same argument shows that the family may be restricted to eligible conditions before certification, which would reduce the Bonferroni penalty; the MIMIC-IV analysis keeps ineligible conditions in the multiplicity family, which is valid but conservative.

3 Inference for Assurance Regions

3.1 Finite-Sample Bounds

Let $\widehat{R}_k$ be the mean of n_k independent losses in $[0, 1]$ with expectation R_k. For binary losses, write $S_k = n_k \widehat{R}_k$. The one-sided exact binomial upper bound is

$$U_k^{\mathrm{CP}} = \begin{cases} Q_{1-\alpha/K}\{\mathrm{Beta}(S_k + 1, n_k - S_k)\}, & S_k < n_k, \\ 1, & S_k = n_k, \end{cases} \tag{3}$$

where Q denotes a quantile (Clopper and Pearson, 1934). For general losses in $[0, 1]$, the Hoeffding bound is

$$U_k^{\mathrm{H}} = \min\left\{1, \widehat{R}_k + \sqrt{\frac{\log(K/\alpha)}{2n_k}}\right\}. \tag{4}$$

Proposition 2 (Finite-sample simultaneous assurance) *Each of the bounds* (3)*, for Bernoulli samples, and* (4)*, for independent bounded samples, satisfies, conditional on* D_0*,*

$$\mathbb{P}\{R_k \le U_k \text{ for every } k \mid D_0\} \ge 1 - \alpha. \tag{5}$$

Consequently, on this event, $\widehat{\mathcal{A}}_\tau \subseteq \mathcal{A}_\tau$ *for every* $\tau \in [0, 1]$ *simultaneously. Dependence between environment samples is allowed.*

Proof An exact one-sided binomial limit fails with probability at most α/K in each coordinate, and for (4) the inequality of Hoeffding (1963) gives the same marginal bound. The union bound yields (5) without independence between coordinates. On that event, $U_k \le \tau$ implies $R_k \le U_k \le \tau$ for every coordinate and every threshold. ■

These constructions are standard and are stated to make the guarantee of the software explicit. For example, $S_k = 0$ gives $U_k^{\mathrm{CP}} = 1 - (\alpha/K)^{1/n_k}$, a positive bound despite no observed errors. Exact Holm uses $p_k = \mathbb{P}\{\mathrm{Bin}(n_k, \tau) \le S_k\}$ and the step-down rule of Holm (1979). These are valid lower-tail tests of $H_{0k} : R_k \ge \tau$ and give finite-sample familywise containment at that fixed tolerance. Exact Holm is an environment-indexed instance of LTT; unlike the bounds above, it does not supply a full upper band. If a finite-sample statement holds conditional on every admissible design, averaging over D_0 preserves its error bound.

3.2 Dependence-Aware Multiplier Band

For paired rows, define $\widehat{\sigma}_k^2 = n^{-1} \sum_i (L_{ik} - \widehat{R}_k)^2$ and

$$T^* = \max_k \left\{ -\frac{1}{\sqrt{n}\,\widehat{\sigma}_k} \sum_{i=1}^{n} \xi_i (L_{ik} - \widehat{R}_k) \right\}, \qquad \xi_i \overset{\text{iid}}{\sim} N(0,1). \tag{6}$$

Let $c^*_{1-\alpha}$ be its conditional $(1-\alpha)$ quantile. The upper band is

$$U_k^{\mathrm{M}} = \min\{1, \widehat{R}_k + c^*_{1-\alpha} \widehat{\sigma}_k / \sqrt{n}\}. \tag{7}$$

The implementation enlarges negative simulated critical values to zero and assigns the bound 1 to empirically constant columns. This avoids invalid studentization but does not correct approximation error for rare events.

Proposition 3 (Fixed-family asymptotic justification) *Under Lemma 1, for fixed K and positive population coordinate variances, the ideal conditional quantile in* (6) *yields* $\liminf_{n\to\infty} \mathbb{P}(R_k \le U_k^{\mathrm{M}} \text{ for all } k \mid D_0) \ge 1 - \alpha$. *For an empirical bootstrap quantile, the number of draws must also increase so that the quantile estimation error vanishes. The statement is pointwise in the risk vector; it is not uniform over designs whose risks or variances approach zero.*

Proof The multivariate central limit theorem applies to the independent bounded loss vectors. The sample covariance converges entrywise to the population covariance, and each sample standard deviation converges to a positive constant. Conditional on the data, the vector in (6) before the maximum is exactly Gaussian with the empirical standardized covariance, so its distribution converges to the Gaussian limit of the standardized mean errors. The maximum of finitely many Gaussian coordinates with nonzero variances has a continuous distribution even if the covariance is singular. Quantile consistency and the continuous mapping theorem give asymptotic coverage, and set containment follows as in Proposition 2. ■

Algorithm 1 summarizes the three procedures. In the applications below, K ranges from 16 to 125 and each family is prespecified, so the fixed-family statement of Proposition 3 is the relevant reference; conditions for increasing K are recorded in Appendix D. Finite-sample validation remains decisive, especially for sparse binary errors. As practical guidance, designs expected to yield fewer than roughly 5–10 errors in any environment should use exact

Algorithm 1: MiSCert estimate of a statistical assurance region

Input: Fixed pipeline g; environment family $\mathcal{E}(D_0) = \{e_1, \dots, e_K\}$; independent assurance sample of size n; tolerance τ; level $1-\alpha$; procedure (exact, Hoeffding, or multiplier); number of multiplier draws B.

Evaluate g in every environment and form the loss matrix $\mathbf{L} \in [0,1]^{n \times K}$

Compute $\widehat{R}_k$ and $\widehat{\sigma}_k$ for $k = 1, \dots, K$

if *procedure is exact (binary losses)* **then**

  $U_k \leftarrow U_k^{\mathrm{CP}}$ from (3)

else if *procedure is Hoeffding* **then**

  $U_k \leftarrow U_k^{\mathrm{H}}$ from (4)

else

  for $b = 1, \dots, B$ **do**

    Draw $\xi_1^{(b)}, \dots, \xi_n^{(b)} \overset{\text{iid}}{\sim} N(0,1)$ and compute T_b^* as in (6)

  $\widehat{c}_{1-\alpha} \leftarrow$ empirical $(1-\alpha)$ quantile of $\{T_b^*\}_{b=1}^B$

  $U_k \leftarrow U_k^{\mathrm{M}}$ from (7)

return $\widehat{\mathcal{A}}_\tau = \{e_k : U_k \le \tau\}$ *and the upper-risk vector* $(U_1, \dots, U_K)$

Clopper–Pearson Bonferroni or exact Holm rather than the multiplier band as the sole basis for certification. Section 4.3 shows that this diagnostic is necessary but not sufficient: with at least 12 expected errors in every environment at $n = 250$, multiplier coverage was 0.810–0.853 for $K = 25$ and 0.610–0.733 for $K = 100$.

3.3 Continuous Environment Families

Suppose a compact continuous family has a scientifically justified global bound $|R(e) - R(e')| \le L_R \operatorname{dist}(e, e')$. Define

$$U^{\mathrm{cont}}(e) = \min\left\{1, \min_{k \le K}[U_k + L_R \operatorname{dist}(e, e_k)]\right\}. \tag{8}$$

On the grid-coverage event, $R(e) \le R(e_k) + L_R \operatorname{dist}(e, e_k) \le U_k + L_R \operatorname{dist}(e, e_k)$ for every k, so (8) covers the risk everywhere and thresholding it gives a continuous inner region. The largest observed slope on the assurance grid is not a valid global bound on L_R, and estimating L_R from the outcomes used for certification invalidates the containment argument. When a data-driven value is unavoidable, an independent pilot sample can propose and fix a conservative smoothness bound before a separate assurance sample is used for the band; if such a bound fails with probability δ, the combined error is at most $\alpha + \delta$. Without such a bound, certification on the finite grid is the defensible output.

The same pilot logic applies to the choice of grid. An exhaustive design with q stress axes and J levels per axis has $K = J^q$ environments, which quickly becomes expensive in data, computation, and interpretation. Our pilot procedure starts with a candidate grid, retains points near a pilot-estimated boundary together with the coordinate extremes, and fixes that family before the independent assurance sample is examined; conditional on the pilot, the exact construction applies without selective-inference corrections. Alternatively, one may

construct a valid band on a fixed superset and report a subset. Reusing the same outcomes to create new environments and then applying an unadjusted band is not justified by either argument. A worked continuous example and a pilot study are given in Appendix B.

3.4 Sample-Size Planning

The Hoeffding bound (4) has additive width at most ϵ when $n \geq \log(K/\alpha)/(2\epsilon^2)$ (Table 13 in Appendix E). This rule controls the width only; it does not guarantee a particular probability of certifying an environment. For binary loss, the exact procedure admits a direct alternative. Because U^{CP} increases with the error count, Clopper–Pearson duality gives the probability that exact Bonferroni certifies an environment with risk R:

$$P_{\mathrm{cert}}(n, R) = \mathbb{P}\{\mathrm{Bin}(n, R) \leq s^*\}, \qquad s^* = \max\{s : \mathbb{P}\{\mathrm{Bin}(n, \tau) \leq s\} \leq \alpha/K\}. \tag{9}$$

The Hoeffding analogue replaces s^* by $\lfloor n\{\tau - \sqrt{\log(K/\alpha)/(2n)}\}\rfloor$. The assurance sample can therefore be planned for a target certification probability at a stated margin $\tau - R$. Because s^* is an integer, P_{cert} is not monotone in n, so we report the smallest n beyond which a target probability is maintained.

4 Simulation Studies

4.1 Design and Evaluation Criteria

All results are generated by the shared implementation with base seed 20260927. Models, designs, and inferential settings are fixed in code, and every table and figure is exported from completed-run outputs. A primary script produces the calibration, adverse-regime, nested-refit, missingness, WDBC, and MIMIC-IV analyses; a supplementary script produces the power study, the exact continuous example, the decomposition of calibration failures, the exact tail calculation, the tolerance–certification profiles, and the MIMIC-IV effective-sample-size summaries, without modifying the primary outputs (Appendix F).

With a known risk vector R, containment is $C = \mathbb{I}\{\widehat{\mathcal{A}}_\tau \subseteq \mathcal{A}_\tau\}$, the assurance-region recall (ARR) is $|\widehat{\mathcal{A}}_\tau \cap \mathcal{A}_\tau|/|\mathcal{A}_\tau|$, and the unsafe-certification proportion (UCP) is $|\widehat{\mathcal{A}}_\tau \setminus \mathcal{A}_\tau|/|\widehat{\mathcal{A}}_\tau|$. In multiple-testing terms, containment is the absence of a familywise error, ARR is the average power, and UCP is the false discovery proportion. We set UCP to zero for an empty certified set and leave ARR undefined when the true region is empty. Wilson intervals describe Monte Carlo uncertainty in coverage and containment frequencies, and mean recalls and paired differences are reported with replication-level Monte Carlo standard errors (MCSEs).

Band coverage, the event $\{R_k \leq U_k \text{ for every } k\}$, and containment answer different questions. Coverage is the event on which Proposition 2 gives containment at every tolerance simultaneously, whereas containment at one tolerance requires only that no environment with $R_k > \tau$ receives $U_k \leq \tau$. A band can therefore miss a coordinate without producing a false certification, either because the missed environment is acceptable or because its bound, although below its risk, is still above τ. We use coverage as the validity criterion for bands and containment for fixed-tolerance procedures such as exact Holm.

Six procedures are compared, always in the same order. Exact Clopper–Pearson Bonferroni (3) gives a finite-sample band for binary loss whose decisions are deterministic given the error counts. Exact Holm gives finite-sample containment at a fixed tolerance but no band.

Hoeffding (4) gives a finite-sample band for any bounded loss and is conservative for small risks. The multiplier band (7) is asymptotically valid for fixed K and uses the dependence across environments. Two normal-theory comparators are computed from the same loss matrix: the Bonferroni normal bound $\widehat{R}_k + z_{1-\alpha/K}\widehat{\sigma}_k/\sqrt{n}$, an asymptotic band that ignores dependence, and the pointwise normal bound $\widehat{R}_k + z_{1-\alpha}\widehat{\sigma}_k/\sqrt{n}$, which has only marginal validity. All procedures use the same $\alpha = .05$, and constant columns receive the bound 1.

4.2 Validity Under Known Risks

To isolate inference, the primary calibration generates binary losses from a Gaussian copula with the 25 marginal risks of the supplied simulation surface, $n = 1000$, latent correlation .55, and $\tau = .34$, using 500 replications and 2,000 multiplier draws per replication. All procedures receive the same loss matrix in each replication. Three adverse designs with 200 replications each then challenge the result: independent Beta losses with concentration four and the means of the main surface; binary risks equally spaced from .002 to .08 at $n = 500$, $K = 25$, and $\tau = .035$, which correspond to roughly 1–40 expected errors per environment; and a $K = 125$ binary family at $n = 1000$ and $\tau = .34$ formed by repeating the main risk values with new copula noise. The last design enlarges the number of hypotheses, not the dimension of a continuous deployment space. Losses remain bounded, and no robustness claim is made for unbounded heavy tails or adversarially chosen environments.

Table 1 and Figure 2 give the results. The finite-sample procedures met their guarantees in every design: exact Bonferroni covered 0.960 in the primary calibration and 0.995 in the rare-binary design, and Hoeffding covered in at least 0.995 of the replications of every design. The Gaussian bands did not. Pointwise normal intervals attained only 0.514 simultaneous coverage, and the multiplier band covered 0.928 in the primary calibration, 0.780 in the rare-binary design, and 0.910 in the $K = 125$ design. Its mean ARR in the primary calibration was 0.297, compared with 0.258 for exact Bonferroni and 0.270 for exact Holm, which retains a finite-sample guarantee at the fixed tolerance. The efficiency gain of the multiplier therefore accompanies a loss of finite-sample validity. Refusing to certify zero-variance columns does not prevent its failure in low-count, nonconstant columns. Hoeffding was very conservative, certifying almost nothing in the bounded-continuous and rare-binary designs. Selecting whichever unadjusted bound yields the most certifications after seeing the assurance losses is not a valid remedy.

Containment stayed near one even when band coverage failed, because band failures in this design rarely produced false certifications. Regenerating the 500 primary replications from the archived seeds (Appendix E), 9 of the 20 exact-Bonferroni coverage failures missed only acceptable environments, and only 1 produced a false certification at $\tau = .34$; for the multiplier, 19 of 34 failures missed only acceptable environments and 1 produced a false certification (Table 14). When an unacceptable environment was missed, its upper bound usually remained between τ and its true risk. Containment at one tolerance therefore says little about the validity of a band.

In the bounded-continuous and rare-binary designs, the Bonferroni normal and multiplier bands have identical coverage; their coverage indicators agree in all 200 replications of both designs, although their certified sets differ. With independent Beta losses, the multiplier critical value (mean 2.87) essentially equals the Bonferroni value 2.88. In the rare-binary

Table 1: Validity and recall under known risks. Coverage is simultaneous band coverage, with 95% Wilson intervals in brackets; exact Holm supplies fixed-tolerance tests rather than a band, so its coverage is not applicable. Panel A: primary calibration (500 replications, $n = 1000$, $K = 25$, containment at $\tau = .34$). Panel B: adverse regimes (200 replications per design); exact binomial bounds do not apply to continuous losses, and containment is evaluated at $\tau = .34$ in the continuous and larger-family designs and at $\tau = .035$ in the rare-binary design.

Panel A. Primary calibration

Method	Guarantee	Coverage	Containment	ARR	UCP
Bonferroni exact	Finite-sample band	0.960 [0.939, 0.974]	0.998 [0.989, 1.000]	0.258	0.0002
Exact Holm	Finite-sample, fixed τ	–	0.998 [0.989, 1.000]	0.270	0.0002
Hoeffding	Finite-sample band	1.000 [0.992, 1.000]	1.000 [0.992, 1.000]	0.138	0.0000
Multiplier	Asymptotic band	0.928 [0.902, 0.948]	0.998 [0.989, 1.000]	0.297	0.0002
Bonferroni normal	Asymptotic band	0.938 [0.913, 0.956]	0.998 [0.989, 1.000]	0.282	0.0002
Pointwise normal	None (marginal)	0.514 [0.470, 0.558]	0.958 [0.937, 0.972]	0.522	0.0042

Panel B. Adverse regimes

Scenario	Method	Coverage [95% CI]	Containment	ARR
Continuous bounded	Hoeffding	1.000 [0.981, 1.000]	1.000	0.008
Continuous bounded	Multiplier	0.925 [0.880, 0.954]	1.000	0.470
Continuous bounded	Bonferroni normal	0.925 [0.880, 0.954]	1.000	0.469
Rare binary	Bonferroni exact	0.995 [0.972, 0.999]	1.000	0.379
Rare binary	Exact Holm	–	1.000	0.379
Rare binary	Hoeffding	1.000 [0.981, 1.000]	1.000	0.000
Rare binary	Multiplier	0.780 [0.718, 0.832]	0.990	0.502
Rare binary	Bonferroni normal	0.780 [0.718, 0.832]	0.995	0.455
Larger binary family	Bonferroni exact	0.960 [0.923, 0.980]	0.995	0.188
Larger binary family	Exact Holm	–	0.995	0.198
Larger binary family	Hoeffding	0.995 [0.972, 0.999]	1.000	0.088
Larger binary family	Multiplier	0.910 [0.862, 0.942]	0.990	0.232
Larger binary family	Bonferroni normal	0.945 [0.904, 0.969]	0.995	0.208

design the multiplier value is smaller (mean 2.82), but failures arise in environments whose standardized errors exceed both values; in 61% of the failing replications, at least one missed environment had at most three observed errors. Both bands rely on the same studentized normal approximation and fail when it fails.

4.3 Power and Sample-Size Planning

Certification depends most directly on the distance between each risk and the tolerance. A factorial power study therefore sets $\tau = .10$ and gives half of the K environments risk $\tau - \Delta$ and half risk $\tau + \Delta$, for $K \in \{25, 100\}$, $n \in \{250, 500, 1000, 2000, 4000, 8000\}$, and $\Delta \in \{.01, .02, .03, .05\}$. Binary losses are generated by the Gaussian copula with latent correlation .55, using 400 replications per cell and 2,000 multiplier draws. The simulated

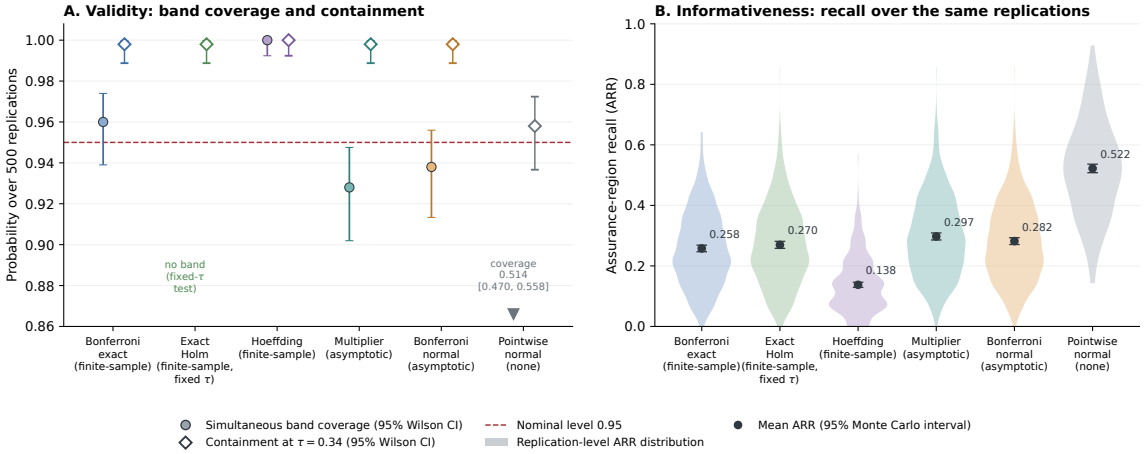


Figure 2: Validity and recall in the primary calibration (500 replications; $n = 1000$, $K = 25$, $\tau = .34$). (A) Simultaneous band coverage (circles) and containment at $\tau = .34$ (diamonds) with 95% Wilson intervals; the dashed line is the nominal level. Exact Holm has no band, and the pointwise-normal coverage lies below the plotted range, as marked. (B) Distribution of ARR over the same replications, with means and 95% Monte Carlo intervals. The corresponding figure for the adverse regimes is in Appendix A.

exact-Bonferroni recall agrees with (9) to within 0.019 in every cell, which checks the simulation and allows (9) to be used for planning without simulation.

Figure 3 shows that the certification probability rises steeply once the width of the bound falls below the margin, and reducing the margin from .05 to .01 moves this transition by more than an order of magnitude in n. At $\tau = .10$, exact Bonferroni needs 422 to 626 assurance observations, for K between 25 and 500, to certify an environment with margin .05 with probability at least 0.8, but 12,179 to 18,235 for margin .01. Exact Holm improved on exact Bonferroni by at most 0.102 in certification probability, and Hoeffding needed far larger samples than either exact procedure. The largest gain of the multiplier, 0.337, occurred in the cell where its band coverage was lowest, 0.610. Its coverage was 0.810–0.853 at $n = 250$ with $K = 25$ and 0.610–0.733 with $K = 100$, although every environment then had at least 12 expected errors, and it remained between 0.907 and 0.955 at $n = 8,000$. Exact Bonferroni coverage was at least 0.948, and containment was at least 0.965 for every procedure (Table 8).

The deficit that remains at $n = 8,000$ for $K = 100$ is not Monte Carlo noise. A studentized binomial mean has a one-sided coverage error of order $n^{-1/2}$ driven by skewness (Hall, 1992), and Wald limits can undercover even at large n (Brown et al., 2001). Simultaneous inference evaluates this error far in the tail, at a per-environment level near α/K. An exact binomial calculation for one environment with risk .09 or .11 shows that the studentized normal upper bound at level α/K fails 8.6–9.6 times as often as nominal at $n = 250$ with $K = 100$, 1.5–1.6 times as often at $n = 8000$ with $K = 100$, and about 1.3 times as often at $n = 8000$ with $K = 25$ (Table 9). Larger families therefore need larger samples before a Gaussian band

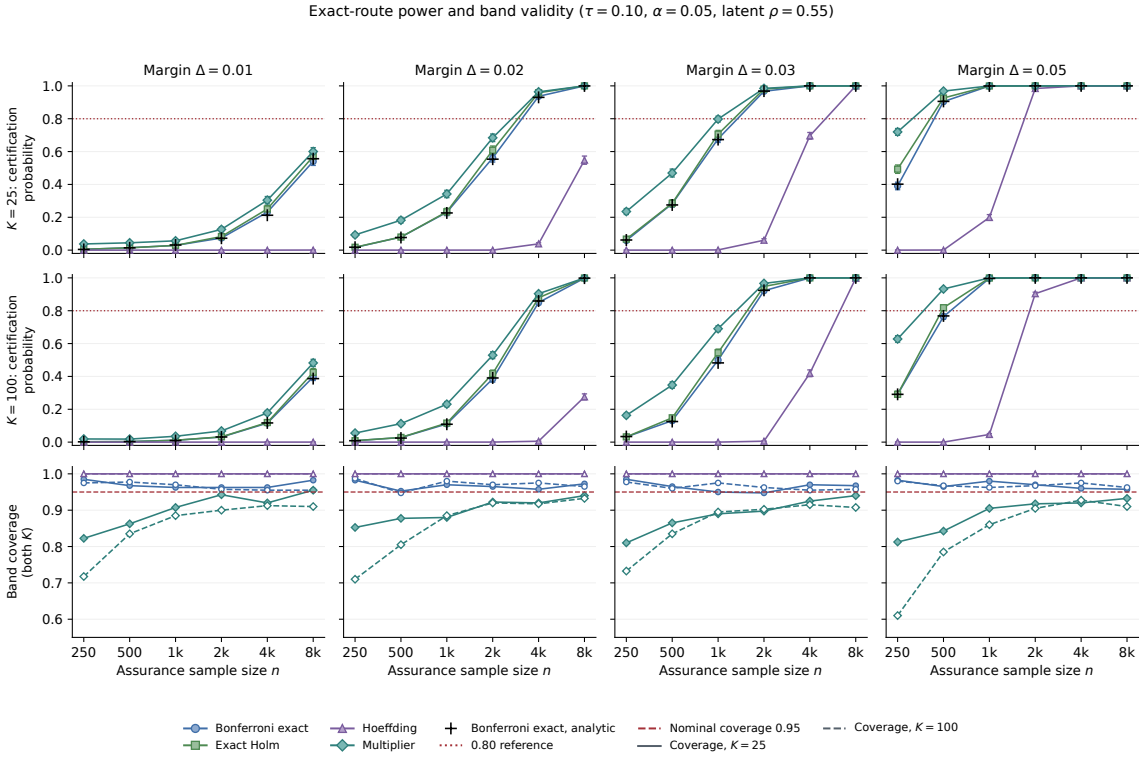


Figure 3: Risk-margin power study ($\tau = .10$, $\alpha = .05$, latent correlation .55, 400 replications per cell). Half of the K environments have risk $\tau - \Delta$ and half $\tau + \Delta$. Top two rows: probability of certifying an environment at risk $\tau - \Delta$ for $K = 25$ and $K = 100$; crosses are the closed-form exact-Bonferroni values from (9), and the dotted line marks 0.8. Bottom row: simultaneous band coverage (solid, $K = 25$; dashed, $K = 100$) with the nominal level 0.95. Holm has no band and is omitted from the bottom row.

attains its nominal familywise level. The multiplier critical value is somewhat smaller than $z_{1-\alpha/K}$ because it uses dependence, but it relies on the same approximation.

Table 2 turns (9) into a planning rule. The exact requirement for probability 0.8 is 0.34–0.40 of the Hoeffding width rule n_{H}, and at n_{H} the Hoeffding procedure itself certifies such an environment with probability only 0.50–0.53: when the width equals the margin, an estimate equal to the true risk lies exactly on the certification boundary. Planning by certification probability rather than by width both reduces the required sample and targets the quantity of practical interest.

Dependence between environments matters less than the margin. In a sensitivity study with 300 replications per setting, varying $n \in \{500, 1000, 5000\}$ at latent correlation .55 and $\rho \in \{.10, .55, .90\}$ at $n = 1000$ (Figure 9 and Table 7 in Appendix A), the paired ARR gain of the multiplier over Bonferroni normal was negligible at low correlation and larger at high correlation. Against exact Bonferroni the gain ranged from 0.016 to 0.092 and was largest at

Table 2: Assurance-sample planning at $\tau = .10$ and $\alpha = .05$ for an environment whose risk is Δ below the tolerance. $n_{\mathrm{H}} = \lceil \log(K/\alpha)/(2\Delta^2) \rceil$ is the Hoeffding width rule. The exact columns give the smallest n from which the exact-Bonferroni certification probability (9) stays at or above the stated level; the last two columns give the certification probability of each procedure at n_{H}.

		Hoeffding	Exact n for $P_{\mathrm{cert}} \geq$		P_{cert} at n_{H}	
K	Δ	width n_{H}	0.8	0.9	Hoeffding	Exact
25	0.01	31,074	12,179	15,138	0.500	0.999
25	0.02	7,769	2,966	3,668	0.502	0.999
25	0.03	3,453	1,285	1,567	0.498	1.000
25	0.05	1,243	422	522	0.526	1.000
100	0.01	38,005	14,991	18,277	0.501	1.000
100	0.02	9,502	3,648	4,418	0.507	1.000
100	0.03	4,223	1,573	1,891	0.501	1.000
100	0.05	1,521	528	616	0.528	1.000
500	0.01	46,052	18,235	21,837	0.500	1.000
500	0.02	11,513	4,431	5,279	0.508	1.000
500	0.03	5,117	1,905	2,258	0.510	1.000
500	0.05	1,843	626	741	0.521	1.000

$\rho = .90$, where the multiplier coverage, 0.967, was also adequate; at $n = 500$ its coverage was 0.903. Strong dependence is therefore the regime in which the efficiency of the multiplier is both largest and best supported.

4.4 Fitted Pipelines and Missingness Mechanisms

The fitted-pipeline experiment uses 200 independent refits. Each generates 1,200 training rows, 400 source-calibration rows for ATC, 1,000 assurance rows, and 20,000 separate reference rows. Ten Gaussian covariates have AR(1) covariance Σ with parameter .35, and outcomes follow

$$\mathbb{P}(Y = 1 \mid X) = \operatorname{expit}\left[1.5\{.75X_0 - .65X_1 + .45X_2 + .55X_0X_1 - .35X_2^2 + .20X_3\}\right]. \quad (10)$$

A fixed normalized direction $\mathbf{v} \propto (.8, -.6, .4, .2, 0, \ldots, 0)$ induces the mean shift $s\,\mathbf{Cv}$, where $\Sigma = \mathbf{CC}^\top$ is the Cholesky factorization. The grid has $m \in \{0, .1, .2, .3\}$ and $s \in \{0, .15, .30, .45\}$, with $\tau = .35$. Feature X_0 remains observed, and the severity m is the missingness rate among the other nine features; imputation uses training means. The two pipelines are standardized logistic regression (LR) and histogram gradient boosting (HGB; 60 iterations, 15 maximum leaves, learning rate .08). The same base rows, label uniforms, and missingness randomness are used across models and environments. Each model is certified separately over its own 16-environment family, so no joint statement over the two models, and hence no model-selection guarantee, is claimed.

The independent reference mean $\widetilde{R}_k$ is itself uncertain. Simultaneous two-sided exact binomial intervals $[a_k, b_k]$ allocate total error .01 over the reference risks of both models within each refit. Cells with $b_k \leq \tau$ are confirmed acceptable, cells with $a_k > \tau$ are confirmed

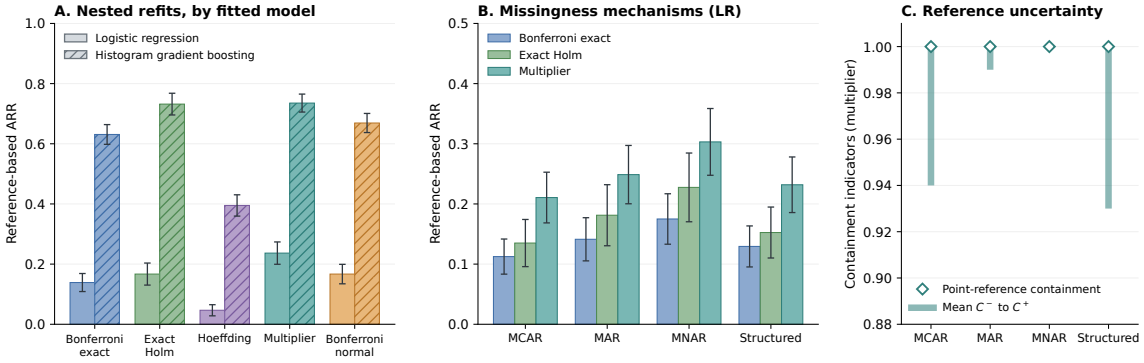


Figure 4: Fitted pipelines and missingness mechanisms. (A) Reference-based ARR across 200 nested refits for logistic regression (plain) and histogram gradient boosting (hatched). (B) Reference-based ARR for three procedures under four missingness mechanisms (100 logistic-regression refits). Error bars in (A) and (B) are approximate 95% intervals based on the replication-level MCSE. (C) Point-reference containment of the multiplier (diamonds) and the mean reference bracket $[C^-, C^+]$ (bars), which is not a confidence interval.

unacceptable, and all others are ambiguous. Then

$$\begin{aligned} C^- &= \mathbb{I}\{\text{all certified cells are confirmed acceptable}\}, \\ C^+ &= \mathbb{I}\{\text{no certified cell is confirmed unacceptable}\} \end{aligned} \tag{11}$$

satisfy $C^- \leq C \leq C^+$ on the reference-coverage event. We report the means of these endpoints as a reference bracket, not as confidence limits for a coverage probability; the reference budget is per refit, not a joint guarantee over refits.

The missingness study uses 100 logistic-regression refits with the same design and compares missing completely at random (MCAR), MAR, missing not at random (MNAR), and structured missingness. MAR depends only on the always-observed X_0, whereas MNAR depends on the latent value of the feature being hidden. Logistic intercepts solve a population Gaussian expectation equation for the target marginal missingness, using 64-node Gaussian quadrature, and are fixed before evaluation rather than normalized with assurance rows. Under each mean shift, the MAR and MNAR models are applied to the covariate centered at the known shifted mean of that environment, so the marginal missingness stays at m in every environment; averaged over environments and refits, the realized fractions differ from their nominal values by at most .0003. Structured missingness makes features 1–3 missing together, each with marginal severity m. Reference intervals allocate error .01 over the four mechanisms within each refit.

Table 3 and Figure 4 show that the fitted pipeline mattered more than the valid procedure that certified it. Exact Bonferroni recovered 0.139 of the reference-acceptable cells for logistic regression and 0.631 for histogram gradient boosting, whereas the two exact procedures differed by about 0.10 or less within either model, and exact Holm and the multiplier had nearly equal recall for HGB. Point-reference containment was 1.000 for every procedure,

Table 3: Fitted pipelines and missingness mechanisms. Reference containment uses the reference means, with 95% Wilson intervals; the reference bracket averages $[C^-, C^+]$ in (11) and reflects ambiguity in the reference risks, not Monte Carlo uncertainty. ARR uses reference means. Panel A: 200 nested refits; LR denotes logistic regression and HGB histogram gradient boosting, each certified over its own 16-environment family. Panel B: multiplier results for 100 logistic-regression refits under each mechanism; ambiguous cells are averaged over the 16 environments per refit.

Panel A. Nested refits

Model	Method	Reference containment [95% CI]	ARR	Reference bracket
LR	Bonferroni exact	1.000 [0.981, 1.000]	0.139	[0.970, 1.000]
LR	Exact Holm	1.000 [0.981, 1.000]	0.167	[0.955, 1.000]
LR	Hoeffding	1.000 [0.981, 1.000]	0.047	[0.995, 1.000]
LR	Multiplier	1.000 [0.981, 1.000]	0.237	[0.940, 1.000]
LR	Bonferroni normal	1.000 [0.981, 1.000]	0.167	[0.960, 1.000]
HGB	Bonferroni exact	1.000 [0.981, 1.000]	0.631	[1.000, 1.000]
HGB	Exact Holm	1.000 [0.981, 1.000]	0.732	[1.000, 1.000]
HGB	Hoeffding	1.000 [0.981, 1.000]	0.395	[1.000, 1.000]
HGB	Multiplier	1.000 [0.981, 1.000]	0.735	[1.000, 1.000]
HGB	Bonferroni normal	1.000 [0.981, 1.000]	0.669	[1.000, 1.000]

Panel B. Missingness mechanisms (multiplier)

Mechanism	Reference containment	Reference bracket	ARR	Ambiguous cells
MCAR	1.000 [0.963, 1.000]	[0.940, 1.000]	0.211	1.83
MAR	1.000 [0.963, 1.000]	[0.990, 1.000]	0.249	0.28
MNAR	1.000 [0.963, 1.000]	[1.000, 1.000]	0.303	0.00
Structured	1.000 [0.963, 1.000]	[0.930, 1.000]	0.232	0.98

model, and mechanism, but this is not proof of perfect containment. For logistic regression, certified cells near the tolerance were sometimes ambiguous under the reference intervals, so the lower endpoint of the bracket ranged from 0.940 to 0.995 across procedures. HGB had no ambiguous cells in these runs, so its favorable recall is a result for this fitted-model regime rather than evidence about difficult near-boundary cases. In the missingness study the lower endpoint ranged from 0.930 to 1.000 and equaled one only for MNAR, the mechanism with no ambiguous cells; these results concern the chosen parameterizations and do not imply that MNAR is generally easier than MCAR.

For comparison, ATC with the maximum class probability as its confidence score, a source-calibrated strict threshold, and tied scores handled jointly predicted the reference risks with mean absolute error 0.031 for logistic regression and 0.030 for histogram gradient boosting (bias -0.012 and -0.014; root mean squared error 0.033 and 0.032). ATC uses no target labels, whereas the certification procedures use labeled assurance losses, so this is a descriptive comparison of a neighboring task rather than a comparison of equal guarantees.

Table 4: WDBC results for separate model-specific 36-environment families. The nominal observed risk is descriptive.

Model	Method	Nominal risk	Certified $\tau = .08$	Certified $\tau = .10$
LR	Bonferroni exact	0.010	15/36	23/36
LR	Exact Holm	0.010	15/36	27/36
LR	Multiplier	0.010	23/36	30/36
LR	Bonferroni normal	0.010	20/36	29/36
Extra Trees	Bonferroni exact	0.035	0/36	9/36
Extra Trees	Exact Holm	0.035	0/36	9/36
Extra Trees	Multiplier	0.035	9/36	19/36
Extra Trees	Bonferroni normal	0.035	2/36	17/36
HGB	Bonferroni exact	0.045	0/36	2/36
HGB	Exact Holm	0.045	0/36	2/36
HGB	Multiplier	0.045	2/36	14/36
HGB	Bonferroni normal	0.045	0/36	12/36

5 Real Data Analyses

5.1 Wisconsin Diagnostic Breast Cancer Data

The WDBC data contain 569 observations. A fixed stratified 65/35 split gives 369 training rows and 200 assurance rows. The fixed models are standardized logistic regression, an Extra Trees ensemble with 500 trees, and histogram gradient boosting. Training means impute missing values, and Gaussian feature noise is scaled by training standard deviations. One noise matrix and one uniform missingness matrix are shared across stress conditions and models. The primary family crosses $m \in \{0, .05, .10, .20, .30, .40\}$ with $s \in \{0, .10, .20, .30, .40, .50\}$, the multiplier uses 5,000 draws, and the tolerances $\tau = .08$ and .10 are illustrative, not medical standards.

Logistic regression has the lowest observed error in this split, and exact Bonferroni certified 15 of its 36 conditions at $\tau = .08$ and 23 at $\tau = .10$ (Table 4 and Figure 5). The counts depend strongly on the procedure and the tolerance. They are separate model-specific simultaneous statements, so choosing a model by comparing them does not confer joint model-selection coverage. The perturbations are synthetic and do not represent observed temporal or inter-hospital shifts, and a single split cannot estimate the true region, ARR, or a coverage probability. Equally spaced 4×4, 6×6, and 10×10 grids for logistic regression are reported in Table 10; certified grid fractions cannot be read as continuous areas without a design measure or a smoothness argument, and no grid was selected retrospectively.

5.2 MIMIC-IV Mortality Prediction

We applied the procedure to credentialed MIMIC-IV v3.1 data (Johnson et al., 2024, 2023) to examine whether it can be carried through on a large electronic health record cohort with approximate temporal grouping and native missingness. We do not establish or quantify covariate or outcome shift across these groups, and neither risk target constitutes a clinically validated deployment rule.

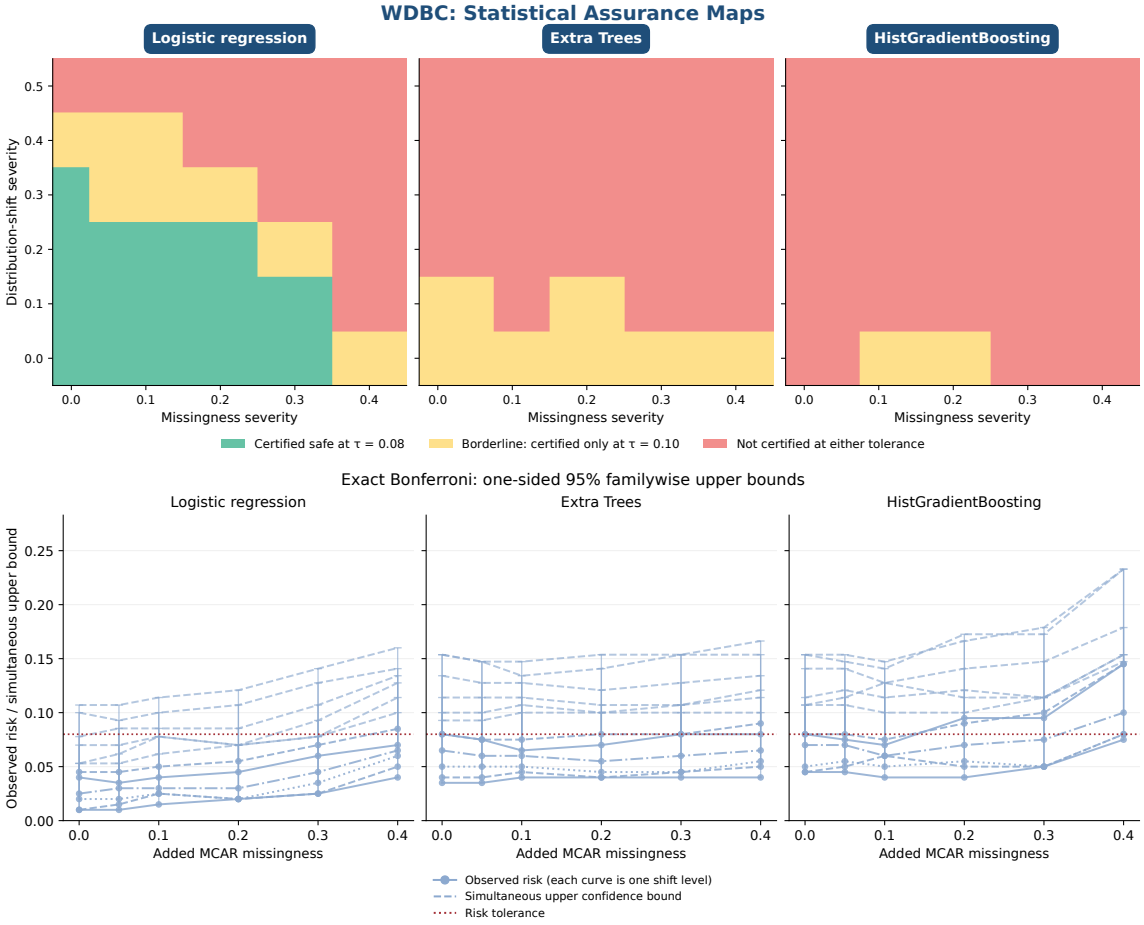


Figure 5: WDBC assurance results for the three models. Top: certification maps over the missingness–shift grid; green cells are certified at $\tau = .08$, yellow cells only at $\tau = .10$, and red cells at neither. Bottom: observed risk (solid) and exact Bonferroni one-sided familywise upper bounds (dashed) against added MCAR missingness, with one curve per shift level; the dotted line marks the tolerance.

5.2.1 Cohort and Design

The raw join contained 546,028 admissions. After retaining valid adult admissions and the first eligible admission per patient, 223,452 patients remained. Admission-aligned temporal intervals placed 106,864 patients in the source cohort and 35,114 in deployment; 81,474 admissions whose assignment was ambiguous under the padded interval rule were excluded. This exclusion is a deterministic function of each patient's own interval metadata and fixed cutoffs and does not couple patients, so the retained observations preserve the independent-row construction of Lemma 1. The source cohort was split into 85,491 fitting observations and a 21,373-observation source holdout. The outcome was in-hospital mortality. Predictors were age, an age-topcoding indicator, gender, admission type, admission location, insurance,

marital status, and race; no laboratory measurements, vital signs, medications, or ICU time series were used, and identifiers, outcome flags, death or discharge times, and length of stay were excluded.

MIMIC-IV shifts dates separately for each patient. We therefore align each admission to the patient anchor year, form approximate admission intervals, pad interval endpoints by one year, and restrict them to the declared v3.1 coverage. The resulting labels are approximate ordering intervals rather than shared calendar periods and should not be read as secular trends. The source period ends by 2016, and the deployment domains are the approximate intervals 2017–2021, 2018–2022, 2019–2022, 2020–2022, 2021–2022, and 2022–2022. The labels may overlap in calendar time, but each patient contributes to one domain. The grouped construction of Section 2.2 accommodates unequal domain sizes, and common-row inference preserves dependence across the model and missingness conditions evaluated on the same patients. Treating patients as independent within a domain ignores shared period effects, such as waves of admissions or changes in practice; the bounds are conditional on that assumption.

Unweighted logistic regression and histogram gradient boosting were fitted on the source training cohort, and an always-survive classifier was retained for comparison. Added MCAR feature missingness took the levels $m \in \{0, .10, .20, .30\}$, with native missingness recorded separately. The three models, six domains, and four missingness levels define one 72-condition family. Before certification we fixed $\alpha = .05$, 5,000 multiplier draws, and a minimum domain size of 200, and all four inferential procedures were applied to the same family.

5.2.2 Misclassification Risk

The tolerance $\tau = .35$ for 0–1 risk was fixed before deployment outcomes were examined. It was chosen as a deliberately loose threshold for executing the procedure, not as a clinically acceptable error rate, and because the source-holdout risks were available when it was finalized, this specification is not a prospective clinical requirement. The source-holdout risks were 0.0184 for logistic regression, 0.0185 for histogram gradient boosting, and 0.0184 for the always-survive classifier. Five domains met the minimum size, giving 60 eligible conditions; the 2022–2022 domain contained only 32 patients, so its 12 conditions were retained as ineligible. All 60 eligible conditions were certified by all four procedures. The largest exact-Bonferroni upper risk was 0.0474, the largest Hoeffding bound was 0.1044, and the multiplier critical value was 2.538 with largest upper bound 0.0465 (Table 5 and Figure 6, top).

This specification exposes a limitation of thresholded 0–1 loss for a rare outcome. At zero added missingness, logistic-regression AUROC ranged from 0.805 to 0.951 across the eligible domains, yet sensitivity at the default 0.5 threshold was zero in four domains and 0.049 in 2019–2022. The always-survive classifier was also certified in every eligible condition because mortality was rare, and this does not depend on the loose value $\tau = .35$: it is certified in every eligible condition at any tolerance above 0.041 (Section 5.3). The certification is internally valid for the specified loss and tolerance but says little about the detection of deaths.

Table 5: MIMIC-IV temporal domains under the misclassification-risk specification. Mortality prevalence, logistic-regression risk, AUROC, and AUPRC are at zero added missingness. "Max U" is the largest exact-Bonferroni upper risk across the three models and four missingness levels in that domain, and the last column gives certified/total conditions. †The 2022–2022 domain is ineligible because $n < 200$; its conditions remain in the family but are not certified, and AUROC and AUPRC are not reported for this one-death domain.

Interval	n	Mort.	LR risk	AUROC	AUPRC	Max U	Cert.
2017–2021	2,312	0.006	0.006	0.805	0.099	0.014	12/12
2018–2022	1,645	0.007	0.007	0.871	0.069	0.020	12/12
2019–2022	29,419	0.037	0.039	0.844	0.176	0.047	12/12
2020–2022	1,282	0.010	0.010	0.914	0.104	0.024	12/12
2021–2022	424	0.012	0.012	0.951	0.173	0.039	12/12
2022–2022	32	0.031	0.031	–	–	–	0/12†

5.2.3 False-Negative Risk

To make the target informative for the minority outcome, a secondary specification bounds the false-negative risk $R_{\mathrm{FN}}(e) = \Pr\{\widehat{Y} = 0 \mid Y = 1, e\}$, analyzed as in Section 2.3. It was introduced after the primary analysis had shown zero sensitivity, so it is prespecified only relative to its own deployment results. Before those results were computed, we fixed $\tau_{\mathrm{FN}} = .20$, corresponding to a sensitivity of at least 0.80; this requirement is interpretable but is not claimed to be a clinical standard. For each learned model, the largest threshold attaining sensitivity 0.80 was selected on the independent source holdout and then fixed: 0.0217 for logistic regression and 0.0227 for histogram gradient boosting, with source sensitivity 0.802 and specificity 0.747 and 0.749, respectively. Because the source holdout is disjoint from the deployment cohort, this calibration does not reuse deployment outcomes. Exact Clopper–Pearson Bonferroni was the primary procedure and exact Holm a secondary check. A minimum of two positive cases per domain was fixed before certification, so the one-death 2022–2022 domain is ineligible.

Positive-case counts in the six domains were 14, 12, 1,092, 13, 5, and 1. Of the 60 eligible conditions, exact Bonferroni certified 3, all for histogram gradient boosting in 2019–2022 at 0%, 10%, and 20% added missingness, with observed sensitivities 0.852, 0.848, and 0.845 and upper false-negative bounds 0.1855, 0.1895, and 0.1925. At 30% missingness the same model had observed sensitivity 0.818 but upper bound 0.218 and was not certified. Logistic regression and the always-survive classifier each certified 0 of 20 conditions, and exact Holm certified the same three conditions and no others (Figure 6, bottom).

The 3-of-60 count describes where the data contain enough positive cases to support a familywise statement rather than ranking the models across domains. For false-negative risk the effective sample size of a domain is its number of deaths n_+. With Bonferroni correction over the full 72-condition family, zero false negatives among n_+ deaths give the smallest attainable upper bound, $1 - (\alpha/K)^{1/n_+}$, so certification at $\tau_{\mathrm{FN}} = .20$ is impossible unless $n_+ \geq 33$. Even zero false negatives among 14 deaths give an upper bound of 0.405, a lower

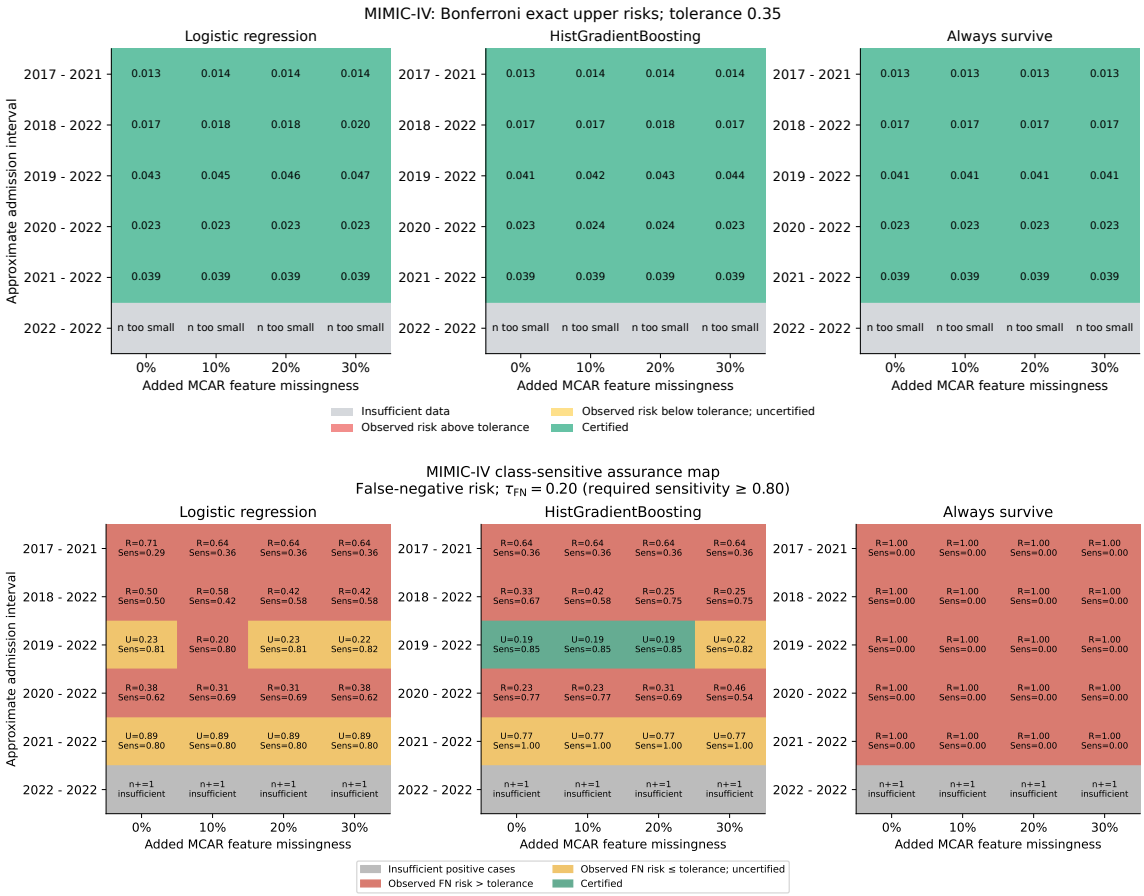


Figure 6: MIMIC-IV assurance maps using exact Clopper–Pearson Bonferroni upper risks. Columns add 0–30% MCAR feature missingness and rows are approximate admission intervals. Top: misclassification risk at $\tau = .35$; the 2022–2022 interval ($n = 32 < 200$) is withheld from certification. Bottom: false-negative risk at $\tau_{\mathrm{FN}} = .20$, equivalent to a required sensitivity of at least 0.80, with thresholds calibrated on the source holdout and then fixed; yellow cells have observed risk at or below the tolerance but an upper bound above it, red cells have observed risk above the tolerance, and gray cells have too few positive cases. Green cells are certified. The maps report statistical support for the prespecified criteria, not clinical safety.

sensitivity bound of only 0.595. Four of the five eligible domains contain 5–14 deaths, so none of their conditions could be certified even with perfect observed sensitivity (Table 6), and the secondary analysis is in effect a single-domain result. In the 2019–2022 domain, with 1,092 deaths, the exact lower bounds separate the two models: histogram gradient boosting has lower sensitivity bound 0.815 at zero added missingness and is certified, whereas logistic regression has 0.766 and is not. Applying (9) to the false-negative rate, a domain whose true sensitivity is 0.90 needs about 230 deaths for an 80% probability of certification, 87

Table 6: Effective sample size of the MIMIC-IV false-negative-risk specification. "Best lower sens." is the exact lower sensitivity bound that zero false negatives would give, $1-(\alpha/K)^{1/n_+}$ with $K = 72$, and certification at $\tau_{\mathrm{FN}} = .20$ requires it to be at least 0.80; observed sensitivities and their exact familywise lower bounds (brackets) are at 0% added missingness, and certified counts are over the four missingness levels.

		Best lower	Can	Sensitivity [lower bound]		Certified	
Interval	Deaths	sens.	certify?	LR	HGB	LR	HGB
2017–2021	14	0.595	no	0.286 [0.031]	0.357 [0.056]	0/4	0/4
2018–2022	12	0.546	no	0.500 [0.105]	0.667 [0.205]	0/4	0/4
2019–2022	1,092	0.993	yes	0.807 [0.766]	0.852 [0.815]	0/4	3/4
2020–2022	13	0.572	no	0.615 [0.184]	0.769 [0.302]	0/4	0/4
2021–2022	5	0.234	no	0.800 [0.111]	1.000 [0.234]	0/4	0/4
2022–2022	1	–	no	–	–	–	–

at true sensitivity 0.95, and 995 at 0.85 (1,205 for a 90% probability). At the mortality prevalence of the four small domains, 0.6–1.2%, 230 deaths correspond to roughly 20,000 to 38,000 admissions per domain, so the number of positive cases should be planned before the analysis.

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The two specifications are reported as separate assurance statements. No adjustment for multiplicity across specifications is claimed, neither is selected post hoc as the better result, and they should not be combined into one familywise statement. The contrast between them is the substantive lesson: inference cannot repair an inadequate loss, threshold, or tolerance, but changing the scientific target changes the assurance question transparently.

5.3 Tolerance–Certification Profiles

A certified count describes a single tolerance. Because a simultaneous band covers every tolerance at once (Proposition 2), the profile $\tau \mapsto |\widehat{\mathcal{A}}_\tau|/K$ computed from one exact-Bonferroni band is covered by the same $1-\alpha$ statement. The display resembles the risk–coverage curves used to evaluate reject-option classifiers (Franc et al., 2023), but it is indexed by environments rather than instances, and it describes sensitivity to the tolerance rather than a device for choosing it. Exact Holm is omitted because its guarantee holds only at a fixed tolerance.

For WDBC (Figure 7A), logistic regression certifies its first cell at $\tau = 0.053$ and all 36 at 0.160, whereas histogram gradient boosting needs 0.233 for all 36. The ordering of the three models is the same at every tolerance up to .15, but both prespecified tolerances fall at the onset or on the steep part of the curves, which is why the WDBC counts change so much between $\tau = .08$ and .10. For MIMIC-IV misclassification risk, every curve, including the always-survive classifier, reaches one below $\tau = .05$. For false-negative risk the 2019–2022 conditions become certified as the tolerance increases from 0.185 to 0.242, while no condition

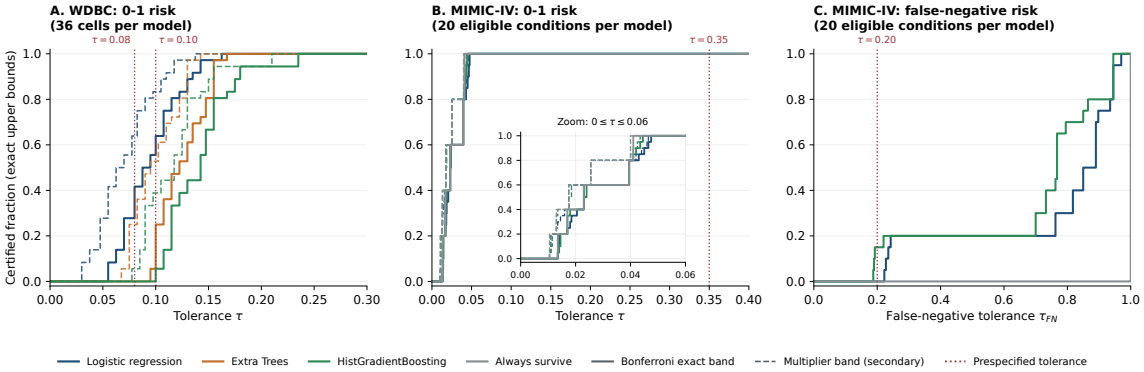


Figure 7: Tolerance–certification profiles: the fraction of each model's family certified at tolerance τ by one simultaneous band, using exact Clopper–Pearson Bonferroni bounds (solid) and, where available, the multiplier band (dashed). Dotted vertical lines mark the prespecified tolerances. (A) WDBC, 36 cells per model. (B) MIMIC-IV misclassification risk, 20 eligible conditions per model; the inset enlarges $0 \leq \tau \leq 0.06$. (C) MIMIC-IV false-negative risk.

in the four small domains is certified below 0.698, a value no practical sensitivity requirement would allow. The conclusion that only the 2019–2022 domain supports a class-sensitive certification therefore holds for every plausible tolerance.

6 Discussion

For binary losses, we recommend exact Clopper–Pearson Bonferroni or exact Holm as the default when the sampling assumptions are defensible. Sparse counts strengthen this recommendation, but a minimum expected error count is not sufficient: the multiplier band undercovered at $n = 250$ with at least 12 expected errors in every environment, more severely for $K = 100$ than for $K = 25$, and it still undercovered at $n = 8,000$ with $K = 100$. The multiplier remains useful as a dependence-aware secondary analysis, particularly under strong dependence, where its gain was largest and its coverage adequate. For nonbinary bounded losses, a bounded-loss inequality is appropriate when conservatism is acceptable. Any use of the multiplier should be accompanied by finite-sample validation in a comparable design. Exact procedures have the further advantage that their decisions are deterministic functions of the error counts, whereas multiplier decisions near the boundary can change with the bootstrap draws (Appendix E).

Because certification depends mainly on the margin between each risk and the tolerance, the assurance sample should be planned for a target certification probability at a stated margin, using (9), rather than for a band width. For class-conditional risks the relevant sample size is the number of cases in the conditioning class; for the MIMIC-IV false-negative target, that means deaths per domain rather than admissions. Grid size is a further constraint. The direct multiplier implementation costs $O(BnK)$ operations, and the conditional-covariance alternative costs $O(nK^2+K^3+BK^2)$ with $O(K^2)$ storage (Appendix C), which reinforces the

recommendation to keep stress axes focused or to screen a larger family with an independent pilot.

Selection and reuse also need care. A fixed band supports all tolerances simultaneously, including one inspected afterward, but this does not justify defining an operational requirement retroactively. Changing models, losses, or families, or choosing the most favorable unadjusted procedure after seeing the assurance losses, requires protection over the enlarged selection problem. Repeated 95% checks are not a time-uniform 95% statement; a simple conservative extension uses fresh assurance data at round t and level $\alpha_t = \alpha/[t(t+1)]$, so that $\sum_{t\geq 1} \alpha_t = \alpha$ and a union bound controls any erroneous certification over all rounds, although this does not give efficient sequential region learning.

The study has several limitations. The inferential components are standard, and no optimality result, minimax recovery rate, or efficient adaptive grid is established. The continuous extension depends on external smoothness, and unknown shifts, causal transportability, unbounded heavy-tailed losses, and adversarial adaptation are outside its scope. Reference-risk uncertainty persists near the tolerance even with 20,000 reference rows. Although the framework accepts any fixed pipeline with bounded losses, the experiments cover a small number of tabular model classes. The primary simulation deliberately isolates inference, and WDBC is small and semisynthetic. The MIMIC-IV analysis uses admission-table predictors from one institutional source, treats patients as independent within approximate temporal domains, adds synthetic MCAR missingness, and does not establish measured covariate or outcome shift; its class-sensitive results come from a single domain and are not a prospective clinical validation. These limitations preclude claims of general deployment readiness or clinical safety.

7 Conclusion

Statistical assurance regions connect a prespecified deployment family to an auditable set-valued report of where a fixed pipeline's risk is supported below a tolerance. Exact binomial and bounded-loss procedures provide this report with finite-sample guarantees, and a Gaussian multiplier band can enlarge it under strong dependence at the cost of undercoverage for binary losses, which persists in large families even at large samples. How much is certified is governed mainly by the risk margin and the assurance sample size, and for the exact procedure this relationship has a closed form that supports planning. The MIMIC-IV analysis shows that a certification is only as informative as its target: a loose misclassification tolerance certified an always-survive classifier, whereas a false-negative target certified 3 of 60 eligible conditions, all in the one domain with enough deaths. Stronger clinical claims require prospectively chosen losses and tolerances, positive-case counts planned for class-sensitive targets, and broader external validation.

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Competing Interests

The authors declare no competing interests.

Data and Code Availability

The source code, analysis notebook, tests, output tables, figures, manuscript sources, and aggregate MIMIC-IV outputs are available in the GitHub repository `https://github.com/dfmaale/statistical-assurance-regions`, and the archived software record is at `https://doi.org/10.5281/zenodo.23241355`. The WDBC data and the simulation surface are supplied with the code. MIMIC-IV is credentialed and is not redistributed; the repository instead contains the fixed v3.1 protocol, cohort-flow, result, and diagnostic tables, missingness summaries, assurance maps, and SHA-256 hashes of the local source tables. Patient-level data, patient-level derivatives, and fitted clinical models are not included.

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Appendix A. Additional Simulation Results

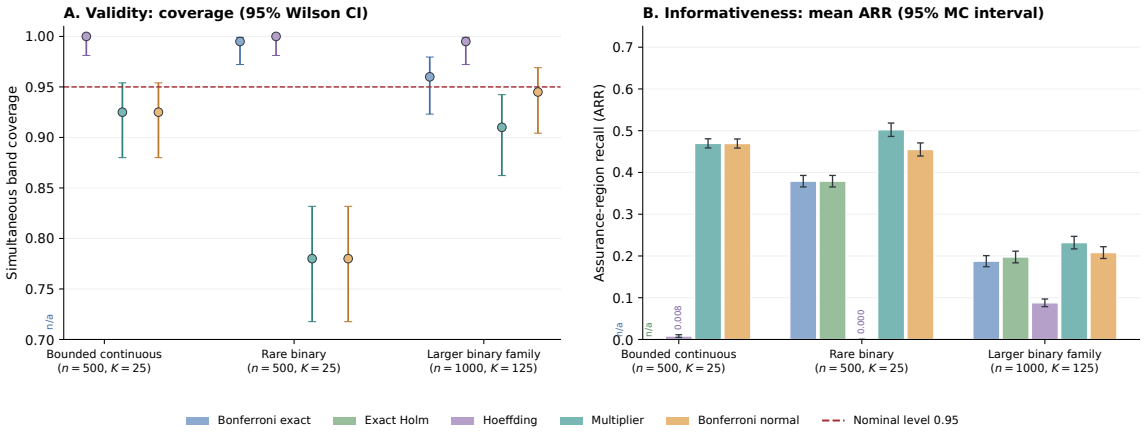


Figure 8: Adverse regimes (200 replications per design). (A) Simultaneous band coverage with 95% Wilson intervals; the dashed line is the nominal level. (B) Mean ARR with 95% Monte Carlo intervals. "n/a" marks procedures that do not apply: exact binomial bounds for continuous losses and, in panel A, exact Holm, which has no band. Bonferroni normal and the multiplier have identical coverage in the first two designs, as explained in Section 4.2.

Table 7: Paired mean ARR gain of the multiplier over each comparator in the sensitivity study. MCSE uses paired replication differences. These efficiency comparisons must be read together with the coverage results.

n	ρ	Comparator	Paired ARR gain	MCSE
500	0.55	Bonferroni normal	0.009	0.0016
1000	0.10	Bonferroni normal	-0.000	0.0008
1000	0.55	Bonferroni normal	0.012	0.0016
1000	0.90	Bonferroni normal	0.066	0.0038
5000	0.55	Bonferroni normal	0.012	0.0017
500	0.55	Bonferroni exact	0.037	0.0029
1000	0.10	Bonferroni exact	0.020	0.0020
1000	0.55	Bonferroni exact	0.036	0.0029
1000	0.90	Bonferroni exact	0.092	0.0041
5000	0.55	Bonferroni exact	0.016	0.0019
500	0.55	Exact Holm	0.035	0.0029
1000	0.10	Exact Holm	0.011	0.0016
1000	0.55	Exact Holm	0.024	0.0023
1000	0.90	Exact Holm	0.080	0.0041
5000	0.55	Exact Holm	0.003	0.0010

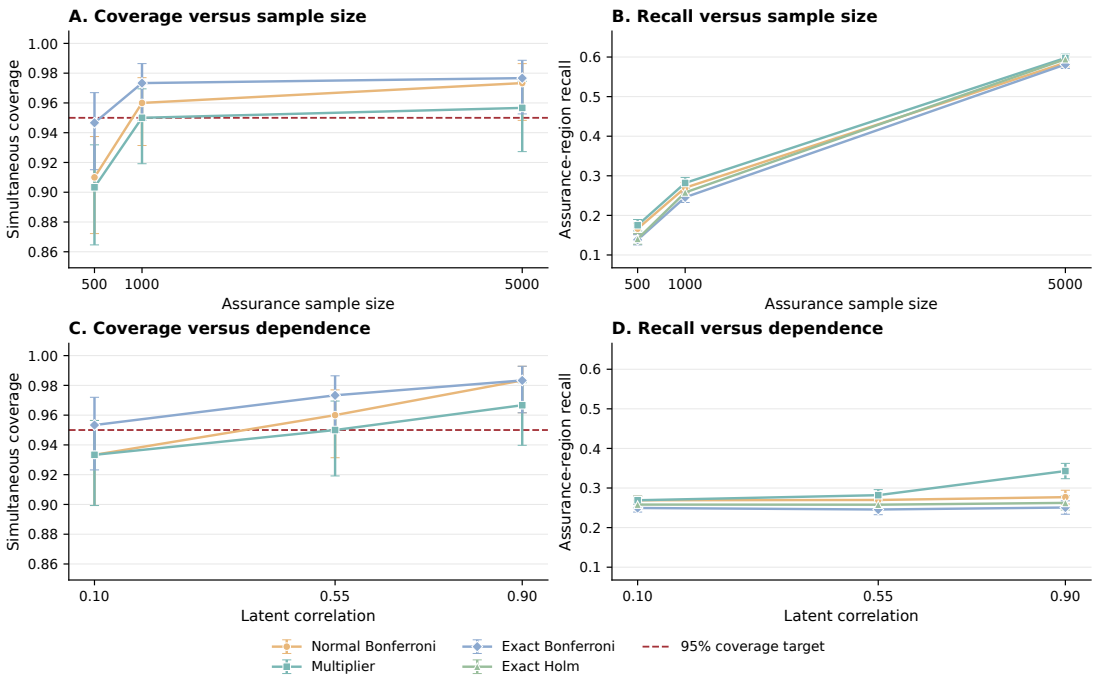


Figure 9: Coverage and recall sensitivity to the assurance sample size and dependence. Latent Gaussian correlation is a generator parameter, not the observed binary-loss correlation. Holm is omitted from the coverage panels.

Table 8: Validity in the power study of Section 4.3: minimum band coverage over the 24 cells of each family size, the minimum at $n = 250$, minimum containment at $\tau = .10$, and maximum UCP. Exact Holm supplies fixed-tolerance tests and has no band.

		Band coverage (min.)		Containment	UCP
Method	K	all n	$n = 250$	(min.)	(max.)
Bonferroni exact	25	0.948	0.983	1.000	0.0000
Bonferroni exact	100	0.948	0.975	0.998	0.0008
Exact Holm	25	–	–	1.000	0.0000
Exact Holm	100	–	–	0.998	0.0008
Hoeffding	25	1.000	1.000	1.000	0.0000
Hoeffding	100	1.000	1.000	1.000	0.0000
Multiplier	25	0.810	0.810	0.983	0.0096
Multiplier	100	0.610	0.610	0.965	0.0126

Table 9: Exact one-sided failure probability of the studentized normal upper bound $\widehat{R}_k + z_{1-\alpha/K}\widehat{\sigma}_k/\sqrt{n}$ for one environment with risk R_k, as a multiple of its nominal value α/K ($\alpha = .05$). Values are computed from the binomial distribution without simulation; a value of 1 would indicate exact calibration.

K	R_k	$n = 250$	$n = 500$	$n = 1,000$	$n = 2,000$	$n = 4,000$	$n = 8,000$	$n = 32,000$
25	0.05	6.54	5.41	2.96	2.29	1.63	1.52	1.24
25	0.09	4.55	2.71	2.50	1.81	1.59	1.31	1.19
25	0.11	4.65	2.22	1.95	1.52	1.50	1.27	1.16
25	0.15	3.98	1.94	1.84	1.38	1.26	1.17	1.13
100	0.05	26.17	11.02	4.39	3.23	2.45	1.80	1.36
100	0.09	8.60	6.42	3.29	2.53	1.72	1.61	1.20
100	0.11	9.60	3.29	2.79	2.32	1.83	1.53	1.24
100	0.15	4.90	3.31	2.23	1.92	1.53	1.31	1.15
500	0.05	45.71	26.10	7.19	5.06	3.12	2.33	1.60
500	0.09	18.63	10.10	4.70	2.90	2.47	1.85	1.38
500	0.11	10.47	9.61	4.46	3.06	2.01	1.75	1.33
500	0.15	12.69	6.52	2.92	2.34	1.70	1.60	1.25

Table 10: Equally spaced WDBC logistic-regression grids.

Grid	Method	K	Certified $\tau = .08$	Fraction
4x4	Multiplier	16	10	0.625
4x4	Bonferroni exact	16	7	0.438
6x6	Multiplier	36	23	0.639
6x6	Bonferroni exact	36	11	0.306
10x10	Multiplier	100	69	0.690
10x10	Bonferroni exact	100	23	0.230

Appendix B. Continuous Example and Independent Pilot

The illustrative surface $R(m, s) = .24 + .12m + .10s + .05ms$ on $[0, .4] \times [0, .6]$ has the valid Euclidean Lipschitz bound $L_R = \sqrt{.15^2 + .12^2}$. We use a 5×5 assurance grid, $\tau = .32$, 200 replications, and an 81×81 mesh for numerical recall and containment diagnostics. The analytical argument for (8), not the mesh, supplies coverage at every point when the grid band and the global bound are valid; the mesh values approximate geometric behavior only (Table 11). Separate exports evaluate half and twice the analytic bound; the half bound is not asserted to be valid.

In the first replication (Figure 10A), fixed in advance rather than selected, exact Bonferroni certified 6 of the 19 acceptable grid points, the Lipschitz extension certified 0.343 of the acceptable mesh area (the multiplier certified 0.379), and no unacceptable mesh point was certified. The extension is conservative near the boundary because the term $L_R \operatorname{dist}(e, e_k)$ is added to bounds that already carry the multiplicity correction.

The pilot example starts with an 11×11 candidate grid on the same rectangle. Using 300 independent pilot rows, it retains points whose pilot risks are within .015 of .32 together with the coordinate extremes, and then certifies the selected family on new assurance rows with exact Bonferroni over 200 replications. This demonstrates a valid split-sample design; it does not establish an optimal adaptive grid or certification of unselected points.

Table 11: Continuous example over 200 replications. Grid coverage is band coverage on the 5×5 grid; mesh containment and recall are evaluated on an 81×81 mesh, where recall is an equal-grid-point area approximation.

Method	Grid coverage	Mesh containment	Mesh recall	Mesh UCP
Bonferroni exact	0.965	0.980	0.404	0.0001
Hoeffding	1.000	1.000	0.185	0.0000
Multiplier	0.940	0.975	0.460	0.0004

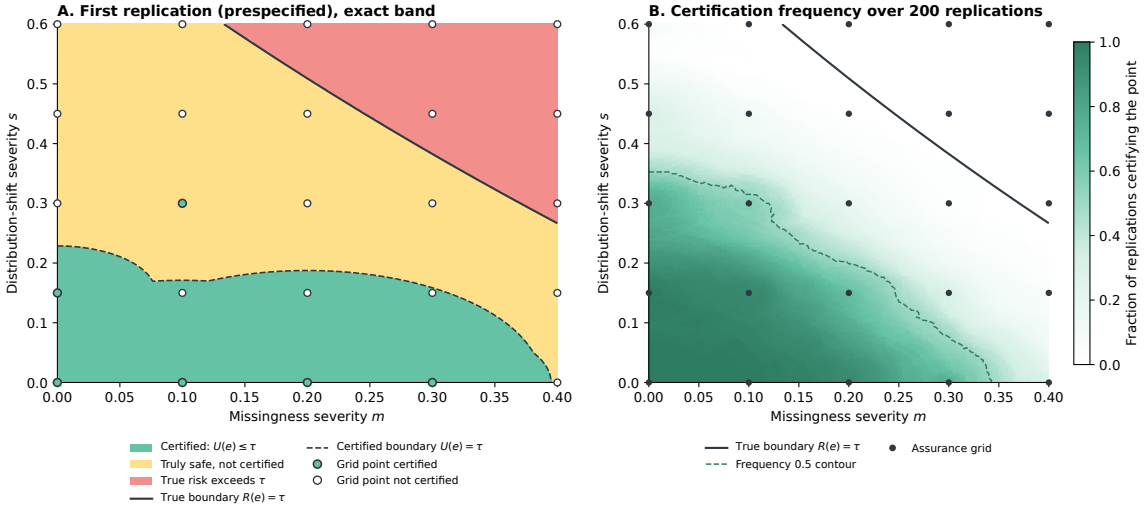


Figure 10: Continuous example with the exact Clopper–Pearson Bonferroni grid band and the Lipschitz extension (8) (5×5 grid, $\tau = .32$). (A) The first of the 200 replications: green is the certified region, yellow is acceptable but not certified, and red has true risk above τ; filled green markers are certified grid points. The certified point at $(m, s) = (0.1, 0.3)$ has an upper bound just below τ, so its certified neighborhood is smaller than its marker. (B) Fraction of replications in which each point is certified, with the 0.5 contour.

Appendix C. Computational Cost and Bootstrap Variability

The direct multiplier sampler requires $O(BnK)$ operations. Sampling from its conditional Gaussian covariance instead costs $O(nK^2 + K^3 + BK^2)$ and stores a $K \times K$ matrix, which can be advantageous for moderate K; the code uses it when $K \leq \min(n, 2000)$ and otherwise uses a chunked row-multiplier sampler. Conditional distributional equivalence of the two samplers is tested with identical-column and independent-column constructions; numerical agreement between samplers does not verify frequentist coverage. The timing study holds the loss matrix fixed and repeats each bootstrap size under 25 independent seeds (Table 12). Timings depend on the machine and numerical libraries and exclude loss generation and model fitting. The critical-value MCSE is the standard error of its mean across seeds, not a guaranteed quantile error bound.

Table 12: Conditional-covariance multiplier timing and variability.

n	K	B	Critical mean	Critical MCSE	Mean seconds
500	25	200	2.819	0.0188	0.0019
500	25	1000	2.795	0.0085	0.0022
500	25	5000	2.796	0.0052	0.0057
1000	25	200	2.806	0.0289	0.0018
1000	25	1000	2.793	0.0089	0.0028
1000	25	5000	2.794	0.0037	0.0077
1000	125	200	3.257	0.0227	0.0104
1000	125	1000	3.213	0.0097	0.0127
1000	125	5000	3.210	0.0035	0.0413

Appendix D. High-Dimensional Gaussian Approximation

For increasing K, the Gaussian-approximation results of Chernozhukov et al. (2013) provide sufficient regimes for bounded coordinates with uniformly positive variances; a familiar conservative regime is $\log^7(Kn)/n \to 0$ under the associated regularity constants. Lemma 1 establishes the independent-vector construction, and bounded losses with a uniform variance lower bound supply the basic moment conditions. Later refinements improve Gaussian and bootstrap approximation rates (Chernozhukov et al., 2022). We do not assert their sharpest rates for the present grouped and stressed-data implementation, and none of them is used as a practical sample-size prescription; the finite-sample experiments assess the approximation directly.

Appendix E. Planning Table, Verification, and Reproduction

Table 13 gives assurance sample sizes sufficient for a Hoeffding additive width of at most the stated margin; Table 2 gives the exact alternative based on certification probability.

Table 13: Conservative assurance-sample planning for a Hoeffding additive width at most the stated margin, with $\alpha = .05$. This is not a guarantee of a particular recall.

K	Margin	Sufficient n
25	0.02	7769
25	0.05	1243
25	0.10	311
100	0.02	9502
100	0.05	1521
100	0.10	381
500	0.02	11513
500	0.05	1843
500	0.10	461

The test suite checks the finite-sample binomial error by enumerating possible counts, the zero-error bound, empirical degeneracy, duplicated columns, the equivalence of the

covariance and row multiplier samplers, unequal samples, overlapping and empty groups, reference boundaries, MAR row locality, ATC ties, and the admission-aligned MIMIC-IV cohort construction. Further tests check the exact certification threshold of (9) against brute-force enumeration of Clopper–Pearson bounds, the closed-form certification probability, the zero-false-negative bound and minimum death count of Section 5.2, the planning search, and the monotonicity of the tolerance–certification profiles. The aggregate-output schema is also exercised on a synthetic data set with the structure of the MIMIC-IV tables, which is used only for testing. The notebook imports the same functions, records configuration and package versions, and generates all manuscript inputs from completed runs; no reported result was selected by changing the seed.

The original full run used Python 3.12.7 with NumPy 1.26.4, and the supplementary analyses used Python 3.14.3 with NumPy 2.5.4, SciPy 1.18.1, pandas 3.0.6, and scikit-learn 1.9.1. Regenerating the primary calibration from its archived seeds in the newer environment reproduced all 500 replications exactly for exact Bonferroni, exact Holm, Hoeffding, Bonferroni normal, and pointwise normal (Table 14). The multiplier reproduced 466 of 500 replications, and its regenerated coverage was 0.932 rather than 0.928. This difference does not reflect a change in the method: the multiplier critical value depends on Gaussian draws produced through an eigendecomposition whose output, for example eigenvector signs, can differ across numerical libraries, so the same seed yields different but identically distributed draws. For the fixed loss matrices of the timing study, the regenerated critical values differed from the archived ones with standard deviation 0.123 at $B = 200$ and 0.021 at $B = 5,000$, the scale of Monte Carlo error. Bitwise reproduction of multiplier decisions requires archiving the critical value or the draws, and all multiplier results in the main text are those of the original archived run.

Table 14: Decomposition of band failures in the regenerated primary calibration (500 replications, $\tau = .34$). "Reproduced" counts replications whose coverage, certified count, and ARR match the archived run. Band failures are split into those that missed only acceptable environments and those that missed at least one unacceptable environment. Containment failures are replications that certified an unacceptable environment at $\tau = .34$.

		Band	Failures that missed		Containment
Method	Reproduced	failures	only safe	an unsafe	failures
Bonferroni exact	500/500	20	9	11	1
Exact Holm	500/500	–	–	–	1
Hoeffding	500/500	0	0	0	0
Multiplier	466/500	34	19	15	1
Bonferroni normal	500/500	31	18	13	1
Pointwise normal	500/500	243	89	154	21

Appendix F. Provenance of Reported Results

The base seed is $s = 20260927$, and $S(a, \ldots)$ denotes the integer seed returned by NumPy's `SeedSequence` with inputs $(s, a, \ldots)$, where r indexes replications and j indexes settings. The primary script `scripts/run_analysis.py` produced, from archived outputs, the primary calibration (losses $S(10, r)$, multiplier $S(11, r)$), the adverse regimes ($S(32, j, r)$, $S(33, j, r)$), the sensitivity study and paired gains ($S(20, j, r)$, $S(21, j, r)$), the independent pilot ($S(80, r)$, $S(81, r)$), the nested refits and ATC predictions ($S(40, r)$–$S(44, r)$, multiplier $S(45, r)$), the missingness study ($S(50, r)$–$S(53, r)$, multiplier $S(54, r)$), the WDBC analyses (split and model seed 42, perturbations s, multiplier $S(60)$), the runtime study ($S(30, n, K)$, $S(31, r)$), and the Hoeffding planning table. The MIMIC-IV misclassification analysis was produced by `scripts/run_mimic.py` or the equivalent notebook call (split, models, and missingness s; multiplier $s + 1$), the false-negative analysis by the notebook (multiplier $s + 8001$), and the operating characteristics by the notebook function `mimic_class_sensitive_operating`, which reuses s without additional bootstrap draws and was checked against the saved thresholds and false-negative counts.

The supplementary script `scripts/run_revision.py` produced the power study (new simulations with seeds $S(90, i_K, i_n, i_\Delta, r)$ and $S(91, i_K, i_n, i_\Delta, r)$), the exact continuous bounds (reusing the losses $S(70, r)$), the decomposition of calibration failures and the adverse-regime critical values (regenerated from the original seeds), and the bootstrap reproduction diagnostics. The exact planning table, the tail calculation, the tolerance–certification profiles, and the MIMIC-IV effective-sample-size summaries are deterministic calculations from saved outputs. Here "archived" means that values come from saved outputs of the original run even when a display was rendered again, "regenerated" means that the analysis was executed again, and reused seeds identify the original random-number construction rather than guaranteeing identical floating-point results across computing environments.